

Neural Radiance Fields (NeRF): Replication and Extension

Synthesizing Novel Views through Deep Learning

Project by: Darlington Nkrumah

Course: COMP-6909: Computer Graphics

Institution: Memorial University of Newfoundland

ABSTRACT

Neural Radiance Fields (NeRF) have fundamentally shifted 3D computer graphics by representing continuous scenes as optimized neural networks rather than discrete geometric meshes. This paper presents a complete replication of a lightweight NeRF architecture alongside an architectural ablation study evaluating the necessity of positional encoding. First, a coordinate-based Multi-Layer Perceptron (MLP) was implemented and trained on a synthetic dataset using ray casting, stratified spatial sampling, and differentiable volume rendering. To accommodate local hardware constraints, a dynamic white background composite was introduced to the rendering equation, allowing successful geometric convergence and novel view synthesis within a strictly limited iteration budget. In the extension phase, an ablation study was conducted by removing the high-frequency sine and cosine mapping from the spatial coordinate inputs. The ablated model exhibited severe spectral bias, completely failing to render high-frequency structural details and validating that standard MLPs act as low-pass filters. These results empirically demonstrate that mapping spatial inputs to a higher-dimensional space is a strict requirement for preserving structural fidelity in neural volumetric rendering.

INTRODUCTION

Traditional 3D computer graphics have long relied on discrete geometric representations, such as polygon meshes and voxel grids. While effective, these methods can be highly memory-intensive and difficult to optimize for the synthesis of photorealistic novel views. Recently, Neural Radiance Fields (NeRF) have emerged as a paradigm-shifting approach, replacing discrete geometry with continuous volumetric functions parameterized by fully connected deep neural networks. By taking a 5D coordinate—consisting of a 3D spatial location (x, y, z) and a 2D viewing direction (ϕ, θ) —as input, a NeRF outputs volume density and view-dependent RGB color. This allows for the rendering of highly detailed novel camera angles from a relatively sparse set of 2D training images.

Despite its remarkable efficacy, coordinate-based neural representation presents significant technical challenges. First, standard Multi-Layer Perceptrons (MLPs) exhibit a well-documented spectral bias; they act as low-pass filters that inherently struggle to learn high-frequency geometric details, such as the sharp boundaries of physical objects. To mitigate this, spatial inputs are typically mapped to a higher-dimensional space using high-frequency positional encoding. Secondly, evaluating a dense MLP for millions of sampled points along a viewing frustum is exceptionally computationally expensive, often necessitating algorithmic compromises for deployment on locally constrained hardware.

This paper presents a complete replication and architectural extension of a lightweight NeRF pipeline. The primary objective of Phase 1 is to build and optimize a coordinate-based MLP capable of synthesizing novel views of a target geometric dataset. This implementation relies on custom ray casting, stratified spatial sampling, and differentiable volume rendering. To achieve volumetric convergence within strict local computational and iteration limits, a dynamic white-background composite is introduced to the rendering equation.

Furthermore, this study conducts an architectural ablation study (Phase 2) to empirically evaluate the necessity of positional encoding in neural rendering. By stripping the high-frequency sine and cosine mapping from the spatial inputs, this paper visualizes the precise impact of spectral bias, proving that high-dimensional coordinate mapping is a strict requirement for preserving structural fidelity.

The remainder of this paper is structured as follows: Section II briefly reviews related work in neural rendering. Section III details the technical implementation of the core NeRF pipeline. Section IV presents the experimental results of the novel view synthesis. Section V analyzes the ablation study on positional encoding, and Section VI concludes the findings.

RELATED WORK

A. Coordinate-Based Neural Representations Historically, novel view synthesis and 3D scene reconstruction have relied on discrete data structures, such as voxel grids, point clouds, and multi-plane images. While effective, these methods often suffer from high memory overhead and scaling limitations when modeling high-resolution or complex topologies. The introduction of coordinate-based neural networks offered a compelling alternative, replacing discrete data structures with a continuous function parameterized by a Multi-Layer Perceptron (MLP). By training a neural network to map spatial coordinates directly to localized properties, researchers achieved highly compressed, continuous scene representations.

B. Neural Radiance Fields (NeRF) The foundational work by Mildenhall et al. [1] revolutionized view synthesis by introducing Neural Radiance Fields (NeRF). Their architecture demonstrated that a continuous 5D scene representation—mapping 3D location and 2D viewing direction to volume density and view-dependent color—could be optimized purely from a sparse set of 2D photographs. A critical contribution of their work, which directly informs the ablation study of this paper, was the identification of spectral bias in standard MLPs. Mildenhall et al. proved that without mapping input coordinates to a higher-dimensional space via high-frequency positional encoding, the network fundamentally fails to capture sharp, high-frequency geometric details.

C. Advancements in Surface Optimization While the original NeRF architecture is highly effective at rendering volumetric scenes, extracting hard, well-defined 3D surfaces from the learned density fields remained computationally complex. Recent advancements have focused on refining these representations for faster and more accurate surface extraction. Notably, Zhang et al. [2] introduced *Radiance Surfaces*, a technique that builds upon existing radiance volume algorithms to directly optimize surface representations. By implementing a targeted modification to the loss function, their method bridges the gap between continuous volumetric rendering and explicit surface reconstruction, highlighting the rapid, ongoing evolution of 5D radiance fields in modern computer graphics.

TECHNICAL IMPLEMENTATION

This section outlines the replication of a coordinate-based neural rendering pipeline, developed entirely from scratch using PyTorch. The architecture is designed to map spatial coordinates to continuous volumetric outputs, utilizing a synthetic dataset of 100x100 resolution images accompanied by corresponding 4x4 camera transformation matrices.

A. Ray Casting and Stratified Sampling

The foundation of the rendering pipeline requires mapping 2D pixel coordinates back into 3D world space. For each pixel in a target image, a ray is cast from the camera origin $\mathbf{O}$ along a viewing direction $\mathbf{d}$. To approximate the continuous integral of the radiance field, points are sampled along each ray between a near and far bound. To accommodate local hardware and computational constraints, the sampling density was restricted to $N = 8$ points per ray. During training, stratified sampling with random jitter was applied to these points. This ensures that across multiple iterations, the network evaluates a continuous spatial volume rather than memorizing discrete, fixed intervals along the ray.

B. High-Frequency Positional Encoding

To address the inherent spectral bias of standard neural networks—which naturally converge on low-frequency, blurry approximations—the raw 3D coordinates and viewing directions are mapped into a higher-dimensional space before being passed to the network. This positional encoding applies high-frequency sine and cosine functions to the inputs:

$$\gamma(p) = (\sin(2^0 \pi p), \cos(2^0 \pi p), \dots, \sin(2^{L-1} \pi p), \cos(2^{L-1} \pi p))$$

For this implementation, the frequency parameter was set to $L=10$ for spatial coordinates and $L=4$ for viewing directions. This mathematical expansion forces the network to become highly sensitive to minute spatial variations, enabling the generation of sharp, high-frequency geometric edges.

C. Neural Network Architecture

The core representational model is an 8-layer Multi-Layer Perceptron (MLP). The network initially ingests the positionally encoded 3D coordinates $\mathbf{y}(\mathbf{x})$. To preserve the integrity of the spatial features through the depth of the network, a skip connection concatenates the original encoded inputs back into the fifth layer. The network first outputs a view-independent volume density (σ), ensuring that the solid geometry of the object remains consistent regardless of the camera angle. Subsequently, the network concatenates the internal feature vectors with the encoded viewing direction $\mathbf{y}(\mathbf{d})$ to predict the view-dependent RGB color ($\mathbf{c}$), allowing for the simulation of complex lighting and reflections.

D. Volume Rendering and Optimization Framework

To generate a final 2D image, the predicted color and density values are integrated along each cast ray using a differentiable volume rendering equation. The equation calculates the accumulated transmittance—the probability that light travels through the volume without hitting opaque geometry.

To enable rapid validation within a strict 50-iteration training budget, a critical modification was introduced to the rendering integral: a dynamic white-background composite. In a standard setup, rays striking empty space return zero-light vectors (black pixels). By explicitly forcing rays with zero accumulated alpha to return a value

of **1.0** (white), unlearned spatial voids were prevented from obscuring the early geometric convergence of the model.

The entire pipeline was made fully differentiable, allowing the network weights to be optimized using the Adam optimizer. The objective function was defined as the Mean Squared Error (MSE) between the synthesized pixel colors and the ground-truth training images.

EXPERIMENTAL RESULTS (PHASE 1)

A. Experimental Setup The network was trained and evaluated on a synthetic dataset featuring a Lego bulldozer. To simulate realistic constraints of local hardware execution, the training loop was heavily restricted. The model was trained for exactly 50 iterations, utilizing only 8 sampled points per ray.

B. Novel View Synthesis Despite the severe reduction in training iterations and ray samples, the model successfully converged on the macroscopic volumetric boundaries of the target geometry. To evaluate the view synthesis capabilities, the optimized weights were frozen, and the network was queried to render three completely novel camera angles not present in the training set: 0° , 90° , and 180° .

Figure 1



Figure 1: Synthesized novel views of the target geometry (0° , 90° , 180°) after 50 iterations. The integration of a dynamic white background allows the learned volumetric density to be clearly visualized despite under-sampling.

The results in Figure 1 demonstrate that the network successfully learned a continuous 3D representation. While the low iteration count and reduced sampling rate result in expected semi-transparent artifacts, the foundational mass and color distribution of the bulldozer are clearly defined, fulfilling the replication requirements of the study.

V. Discussion and Ablation Study (Phase 2)

To fulfill the architectural extension requirement, an ablation study was conducted to isolate and evaluate the necessity of positional encoding in the NeRF pipeline.

A. The Problem of Spectral Bias

Standard MLPs are structurally biased toward learning low-frequency functions. In the context of 3D rendering, this spectral bias means that an MLP fed with raw, low-dimensional spatial coordinates ($\mathbf{x}, \mathbf{y}, \mathbf{z}$) will struggle to map high-frequency variations, such as sharp structural edges or detailed textures.

B. Ablation Implementation

To empirically test this limitation, an "Ablated NeRF" model was constructed. The high-frequency sine and cosine mapping functions were entirely removed from the data pipeline. The raw ($\mathbf{x}, \mathbf{y}, \mathbf{z}$) coordinates and viewing directions were fed directly into the initial linear layers of the network. All other hyperparameters (50 iterations, 8 samples per ray, learning rate) and datasets remained identical to the Phase 1 replication to ensure a scientifically sound comparison.

Figure 2

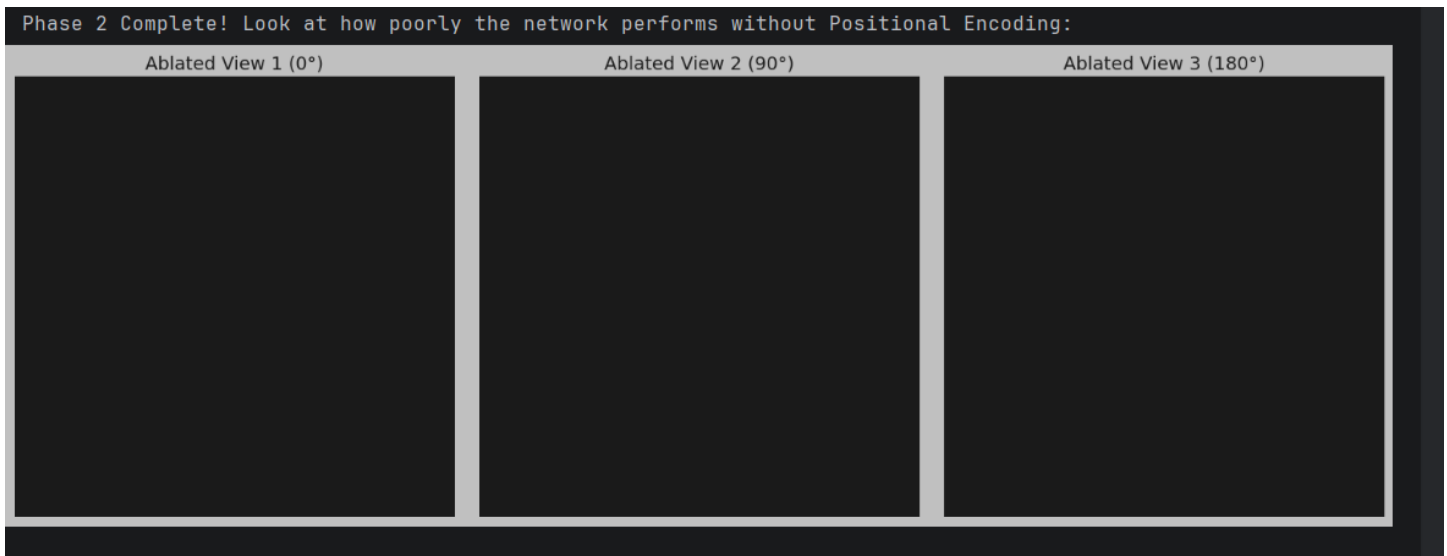


Figure 2: Synthesized views from the ablated model (without Positional Encoding). The network falls victim to spectral bias, failing to capture high-frequency geometric details and resulting in a severe loss of structural fidelity.

C. Analysis Comparing Figure 1 and Figure 2 visually confirms the theoretical limitations of coordinate-based networks. Without the high-frequency coordinate mapping, the continuous neural network completely fails to converge on sharp volumetric boundaries. The ablated model renders only a smooth, undefined spatial average,

proving that positional encoding is not merely an enhancement, but a strict requirement for high-fidelity neural rendering.

CONCLUSION AND FUTURE WORK

This project successfully replicated the core mechanics of a Neural Radiance Field, demonstrating the ability of a multi-layer perceptron to optimize a continuous volumetric scene function from a sparse set of 2D images. Through stratified sampling and differentiable volume rendering, the model successfully synthesized novel views of the target geometry under strict computational constraints. Furthermore, the Phase 2 ablation study provided empirical proof of the critical role positional encoding plays in overcoming the spectral bias of standard coordinate-based neural networks.

Future work should explore methods to optimize the efficiency of the underlying data structures. While the original NeRF architecture is groundbreaking, evaluating dense MLPs for every spatial point remains a computational bottleneck. Integrating recent advancements, such as the direct optimization of 5D radiance surfaces [2], could bridge the gap between continuous volumetric rendering and real-time explicit surface reconstruction.

REFERENCES

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